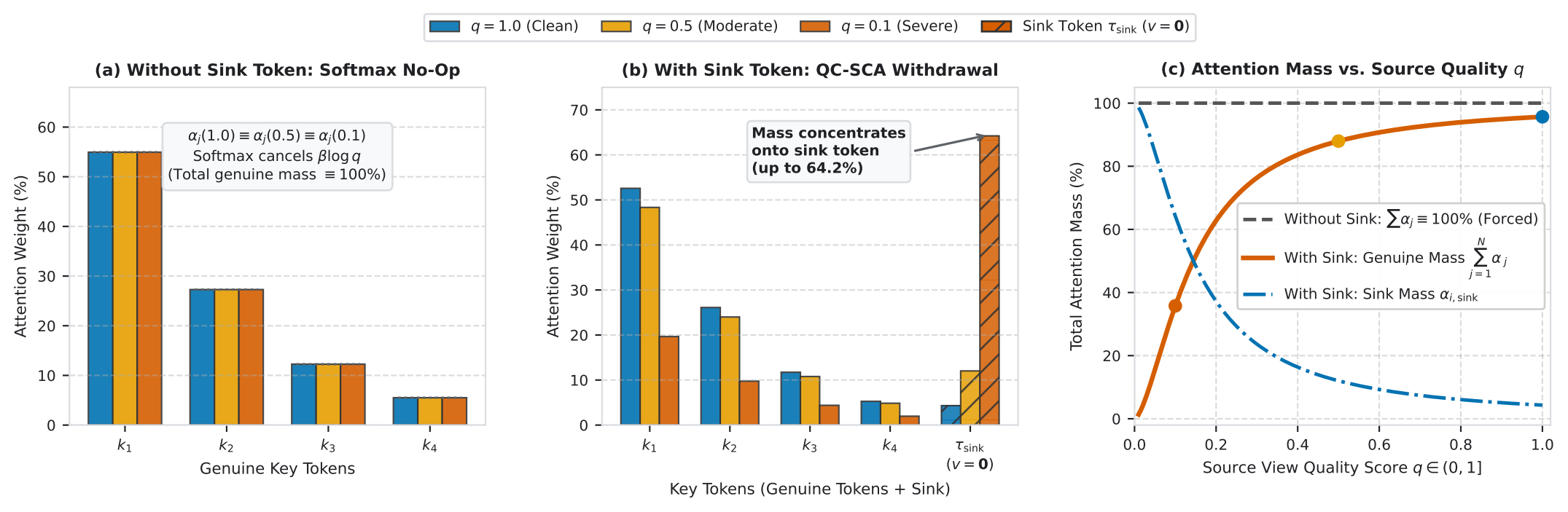
$q = 1.0$ (Clean)
$q = 0.5$ (Moderate)
$q = 0.1$ (Severe)
Sink Token τ_{sink} ($v = \mathbf{0}$)
(a) Without Sink Token: Softmax No-Op
$\alpha_j(1.0) \equiv \alpha_j(0.5) \equiv \alpha_j(0.1)$
Softmax cancels $\beta \log q$
(Total genuine mass $\equiv 100\%$)
Attention Weight (%)
Genuine Key Tokens
(b) With Sink Token: QC-SCA Withdrawal
Mass concentrates
onto sink token
(up to 64.2%)
Attention Weight (%)
Key Tokens (Genuine Tokens + Sink)
(c) Attention Mass vs. Source Quality q
Without Sink: $\sum \alpha_j \equiv 100\%$ (Forced)
With Sink: Genuine Mass $\sum_{j=1}^{N} \alpha_j$
With Sink: Sink Mass $\alpha_{i,\text{sink}}$
Total Attention Mass (%)
Source View Quality Score $q \in (0, 1]$